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AWS EC2, S3 and IAM Assignment

1. Launch and Configure an EC2 Instance

For this task, I created an EC2 instance in AWS and configured the basic settings required to run it securely.

Steps I followed:

1. I first logged in to the **AWS Management Console** and opened the **EC2 service**.
2. From the EC2 dashboard, I selected the option to **Launch Instance**.
3. I gave the instance a suitable name so that I could identify it easily.
4. I selected **Amazon Linux** as the operating system for the instance.
5. I chose a small instance type that was sufficient for performing this assignment.
6. I created a **key pair** so that I could securely connect to the instance if required.
7. I configured a **security group** and allowed SSH access through port 22.
8. For better security, I restricted SSH access to **My IP** instead of allowing access from all IP addresses.
9. I configured the required storage for the instance and enabled **encryption**.
10. After checking all the settings, I launched the EC2 instance.
11. I waited for the instance to start and checked whether its status changed to **Running**.
12. Finally, I verified that the instance was created successfully and captured screenshots of the important settings for my assignment.

Result

The EC2 instance was successfully created and configured with the required operating system, storage, security group, key pair, and access settings.

2. Create an S3 Bucket with a Lifecycle Policy

For the second task, I created an S3 bucket to store objects and configured a lifecycle rule to automatically manage older files.

Steps I followed:

1. I opened the **S3 service** from the AWS Management Console.
2. I selected **Create Bucket** and provided a unique name for the bucket.
3. I selected the required AWS Region for the bucket.
4. I kept **Block all public access** enabled because the files did not need to be publicly accessible.
5. I enabled **Bucket Versioning** so that older versions of objects could be retained.
6. I enabled **Server-Side Encryption** to provide protection for the data stored in the bucket.
7. After checking the settings, I created the S3 bucket.

8. I opened the bucket and went to the **Management** section.
9. I selected the option to create a **Lifecycle Rule**.
10. I gave the lifecycle rule a suitable name and configured it to apply to the objects in the bucket.
11. I added storage transitions so that older files could automatically be moved to more cost-effective storage classes.
12. I configured the required transition periods, such as moving objects to **Standard-IA after 30 days** and to an archival storage class after a longer period.
13. I also configured an expiration period for deleting old objects according to the assignment requirement.
14. After completing the configuration, I saved the lifecycle rule.
15. Finally, I checked that the lifecycle rule was active and captured screenshots of the S3 bucket and its configuration.

Result

The S3 bucket was successfully created with restricted public access, versioning, encryption, and a lifecycle policy for automatically managing older files.